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Sequence to
sequence models

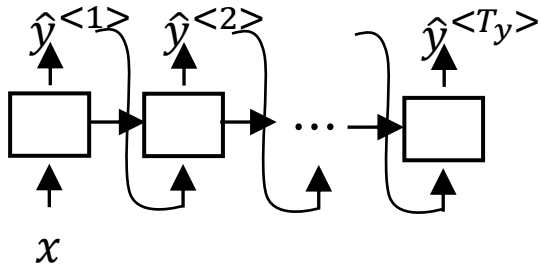
Transformers
Intuition

Transformers Motivation

Increased complexity,
sequential

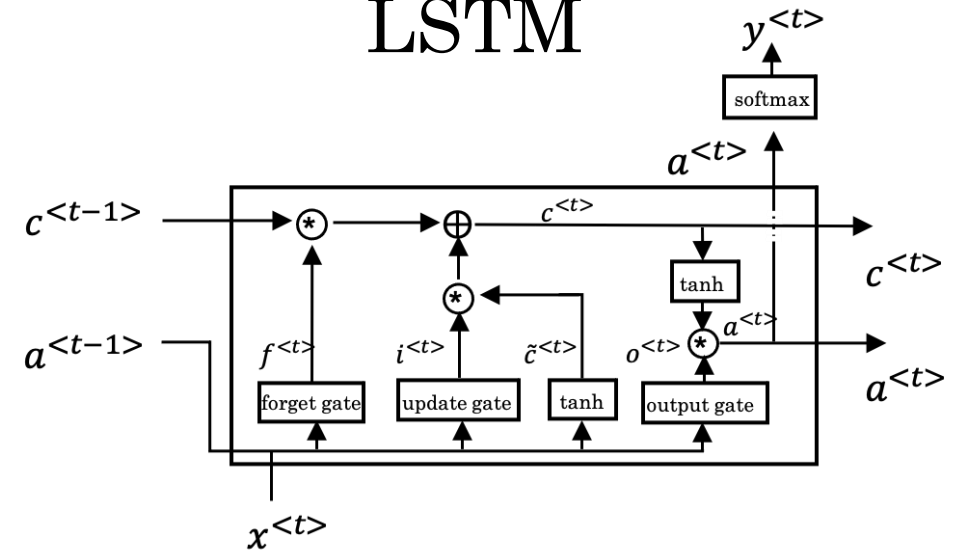


RNN



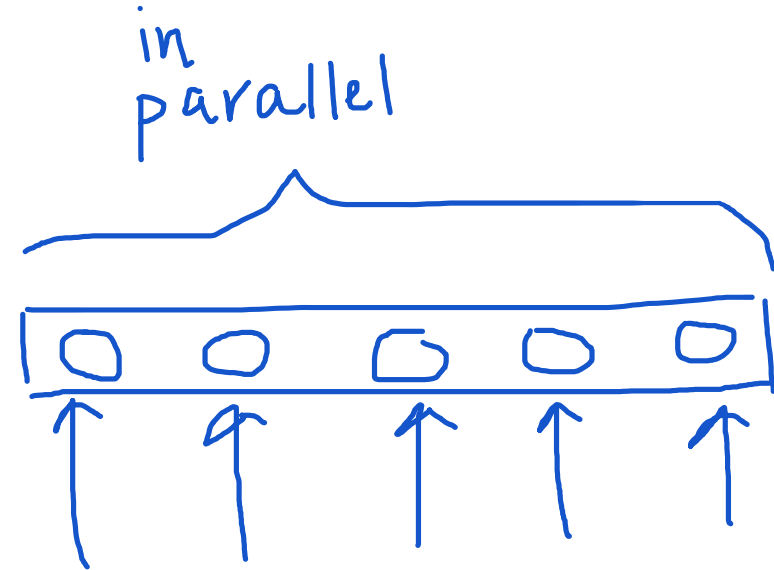
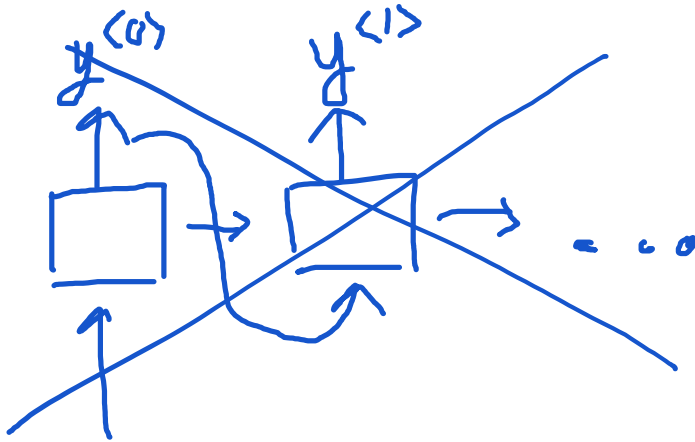
GRU

LSTM



Transformers Intuition

- Attention + CNN
 - Self-Attention
 - Multi-Head Attention





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Sequence to sequence models

Self-Attention

Self-Attention Intuition

$A(q, K, V)$ = attention-based vector representation of a word

↪ calculate for each word

RNN Attention

$$\alpha^{<\cancel{t}, t'>} = \frac{\exp(e^{<t, t'>})}{\sum_{t'=1}^{T^x} \exp(e^{<t, t'>})}$$

$x^{<1>}$
Jane

$x^{<2>}$
visite

$x^{<3>}$
l'Afrique

$x^{<4>}$
en

$x^{<5>}$
septembre

Transformers Attention

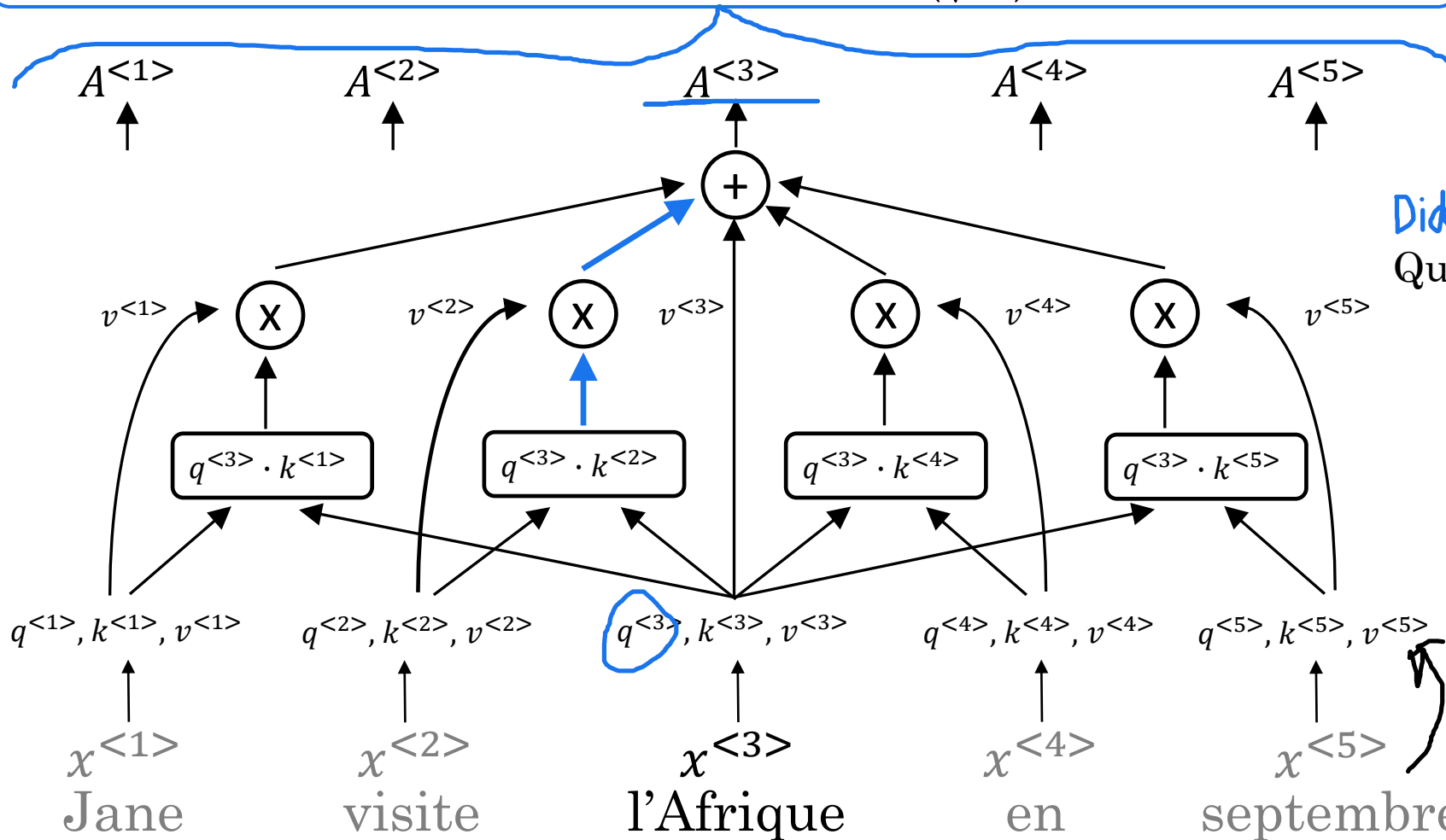
$$A(q, K, V) = \sum_i \frac{\exp(e^{<q \cdot k^{<i>}>})}{\sum_j \exp(e^{<q \cdot k^{<j>}>})} v^{<i>}$$

Self-Attention

$$A(q, K, V) = \sum_i \frac{\exp(e^{< q \cdot k^{<i>} >})}{\sum_j \exp(e^{< q \cdot k^{<j>} >})} v^{<i>}$$

softmax

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Did what?

Query (Q)

Key (K)

Value (V)

$q^{<1>}$

$k^{<1>}$

$v^{<1>}$

$q^{<2>}$

$k^{<2>}$

$v^{<2>}$

$q^{<3>}$

$k^{<3>}$

$v^{<3>}$

$q^{<4>}$

$k^{<4>}$

$v^{<4>}$

$q^{<5>}$

$k^{<5>}$

$v^{<5>}$

What's happening there?

person
action
visit

W^Q, W^K, W^V

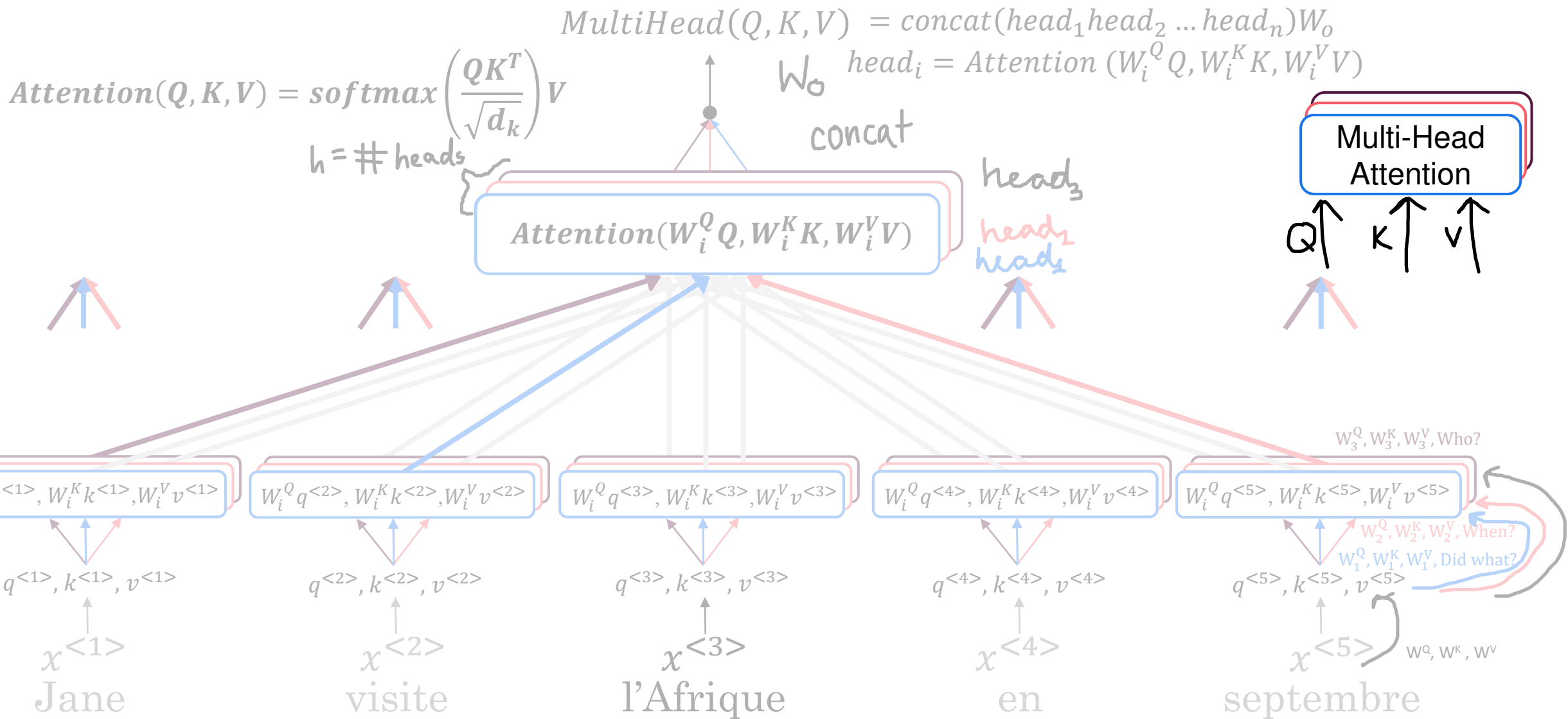


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Sequence to sequence models

Multi-Head Attention

Multi-Head Attention





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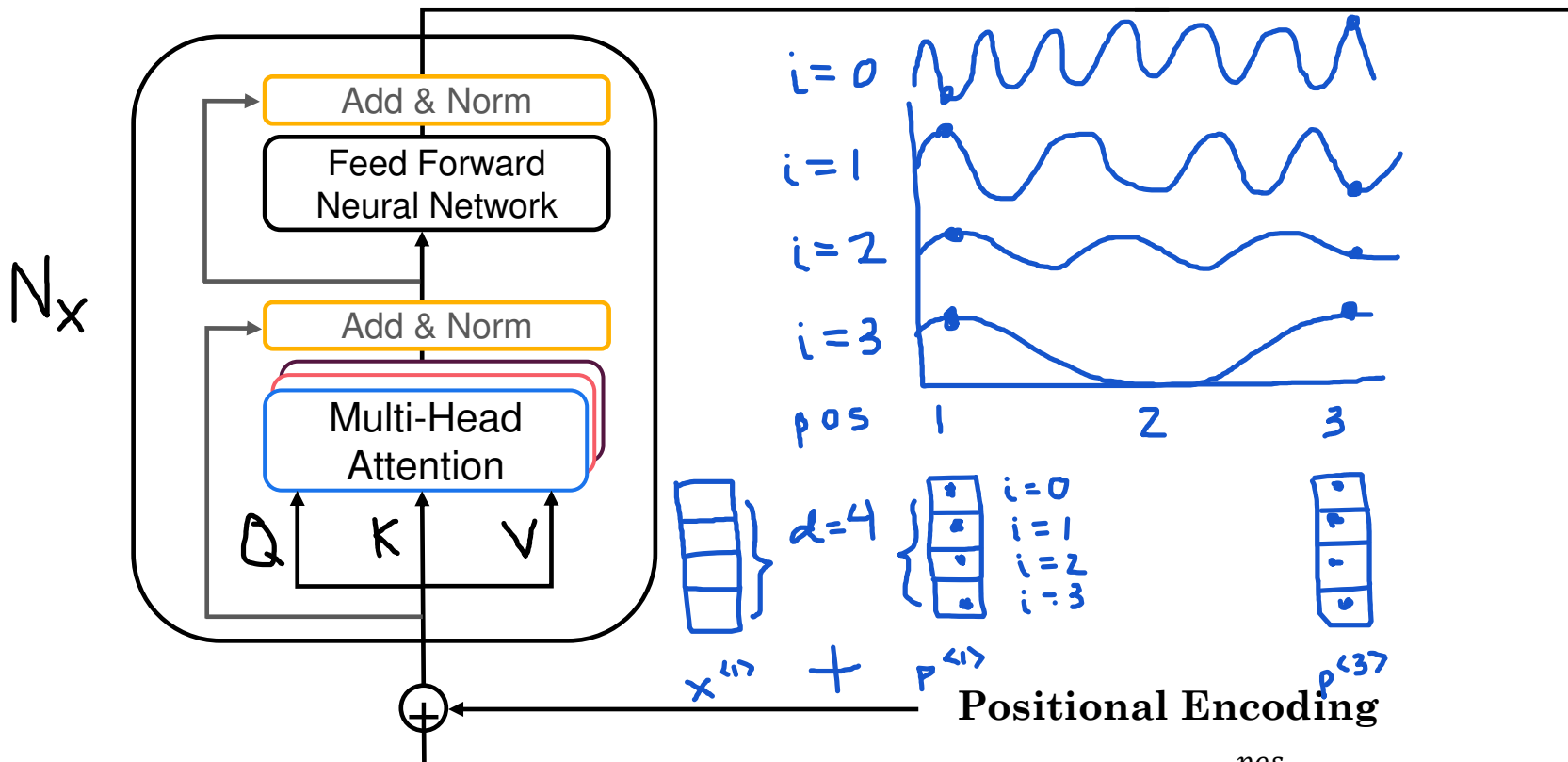
Sequence to sequence models

Transformers

Transformer Details

<SOS> Jane visits Africa in September <EOS>

Encoder

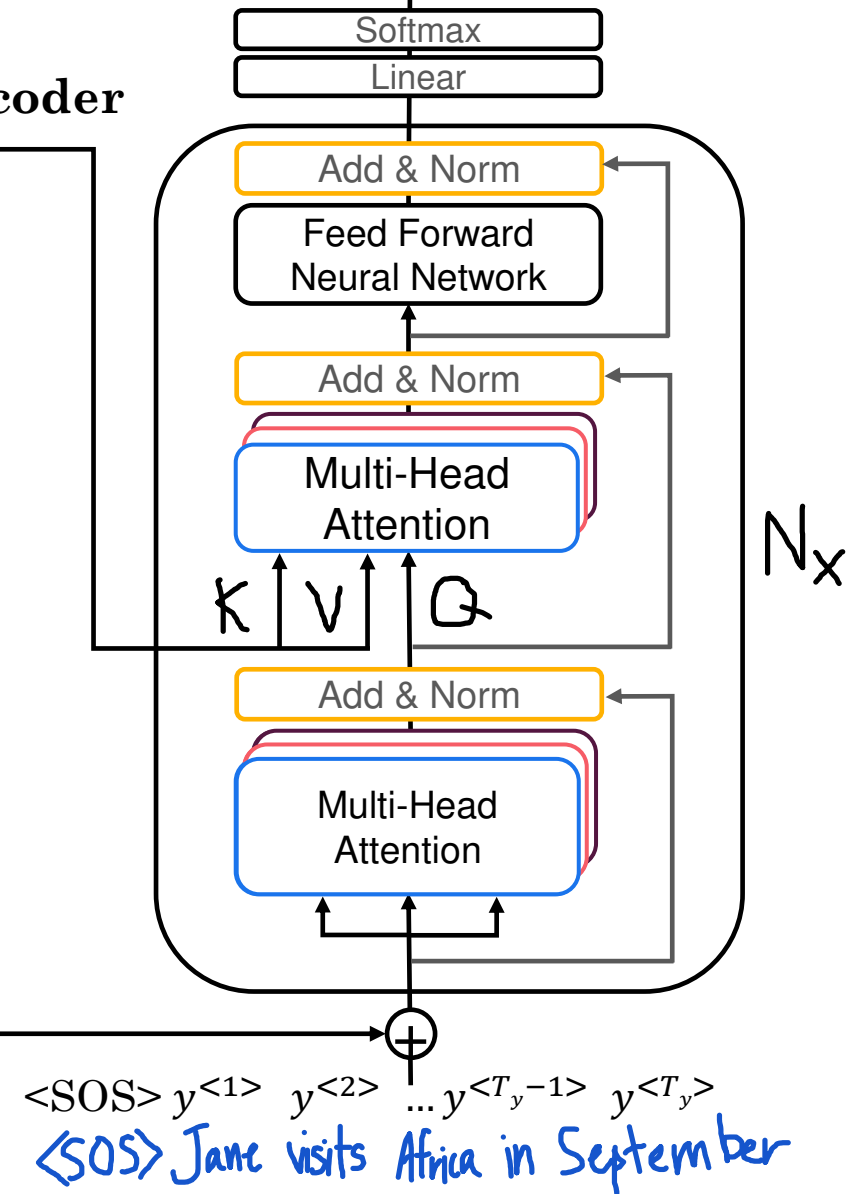


$\langle \text{SOS} \rangle x^{<1>} x^{<2>} \dots x^{<T_x-1>} x^{<T_x>} \langle \text{EOS} \rangle$
Jane visite l'Afrique en septembre

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{1000^{\frac{2i}{d}}}\right)$$

$$PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{1000^{\frac{2i}{d}}}\right)$$

Decoder



$\langle \text{SOS} \rangle y^{<1>} y^{<2>} \dots y^{<T_y-1>} y^{<T_y>}$
 $\langle \text{SOS} \rangle$ Jane visits Africa in September